

GFCI, AFCI and the Protection Devices

Three devices, three completely different dangers. Match the device to the danger and these stop being guessable.

Breaker	Stops overcurrent. Too many amps for the wire.
GFCI	Stops electrocution. It operates by sensing current imbalance within the circuit — current leaving that does not come back is going somewhere it should not, possibly through a person.
AFCI	Stops fire. An arc can start a fire while drawing less current than normal, so a plain breaker cannot see it at all.

YOU HAVE MISSED THIS

A device that shuts off a circuit immediately if it senses an arc is called a/an:

Arc Fault Circuit Interrupter (AFCI).

The ungrounded outlet problem

An ungrounded outlet can be replaced by a GFCI installed upstream of the outlet. The GFCI does not create a ground; it watches for imbalance, which works whether or not a ground is present. That is why it is the accepted remedy in an older house where running a ground wire is impractical.

Where GFCI protection is required

Think water and think outdoors: kitchens, bathrooms, garages, basements, crawl spaces, exteriors, and around pools. The exam tests the edge of the list rather than the middle of it.

YOU HAVE MISSED THIS

According to the NEC, GFCIs must be installed near all of the following EXCEPT:

Attic access opening.

An attic access is dry and is not a wet or damp location, so it falls outside the requirement. On an EXCEPT question, find the item that does not share the quality the others have in common — here, exposure to moisture.

One more device that trips and resets

YOU HAVE MISSED THIS

A properly installed garbage disposal that makes no sound when turned on can sometimes be activated by:

Pressing the thermal overload switch.

A silent disposal usually means its internal thermal overload has tripped, not that the circuit is dead. The reset button sits on the underside of the unit. Silence points to the appliance's own protection; a dead circuit

would take other things with it.

Say it back

Cover the page. Answer each one out loud.

1. Breaker, GFCI, AFCI — which danger does each one stop?
2. How does a GFCI actually detect a problem?
3. What may replace an ungrounded outlet, and where must it go?
4. Why is an attic access opening the exception on the GFCI list?
5. A disposal makes no sound. What do you try first?